Computing Theory and Practices (2026)
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M2U2

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M2U2 Plan

1. Quantum Operations
2. Single-Qubit Gates: X, Y, Z and H
3. Phase Gates
4. CNOT and Multi-Qubit Gates
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9. Deutsch Algorithm
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15. Tutorial: Algorithms
16. Tutorial: Quantum Programming

Learning Outcomes

After completing this lecture, you will be able to

- Explain the concept of a quantum operation.
- Distinguish between unitary evolution and general quantum operations.
- Describe the Kraus operator representation.
- Understand how noise affects quantum states.
- Identify applications of quantum operations in quantum computing.

Why Quantum Operations?

An isolated quantum system evolves according to

$$\rho' = U\rho U^\dagger.$$

However, real quantum systems interact with their surroundings.

Examples include

- Thermal noise
- Decoherence
- Measurement
- Imperfect quantum gates

A more general mathematical description is therefore required.

What is a Quantum Operation?

A quantum operation is a mathematical rule that transforms one quantum state into another.

It acts on density matrices as

$$\boxed{\rho \longrightarrow \mathcal{E}(\rho)}$$

A valid quantum operation must satisfy

- Linearity
- Complete positivity
- Trace preservation

Kraus Representation

Any quantum operation can be written as

$$\boxed{\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger}$$

where E_k are called **Kraus operators**.

For a trace-preserving operation,

$$\sum_k E_k^\dagger E_k = I.$$

Examples of Quantum Operations

Common quantum operations include

- Unitary evolution
- Projective measurement
- Bit-flip channel
- Phase-flip channel
- Depolarizing channel
- Amplitude damping

Each operation changes the quantum state in a different way.

Unitary Evolution vs Noise

Unitary Evolution	**Quantum Noise**
Closed system	Open system
Reversible	Often irreversible
Pure states remain pure	Pure states may become mixed
No information loss	Information may leak to the environment

Applications

Quantum operations are essential for

- Quantum error correction
- Quantum communication
- Noise modelling
- Quantum process tomography
- Quantum hardware characterization
- Quantum algorithm simulation

They provide a realistic description of practical quantum devices.

Worked Example

Suppose a qubit undergoes a perfect Hadamard gate.

This is described by

$$\rho' = H\rho H^\dagger.$$

If the same qubit experiences environmental noise after the gate, its evolution is more accurately described by

$$\rho' = \sum_k E_k \rho E_k^\dagger.$$

The Kraus operators account for the effects of the environment.

Summary and Concept Check

Summary

- Quantum operations generalize unitary evolution.
- They describe both closed and open quantum systems.
- Kraus operators provide a compact mathematical representation.
- Quantum operations model measurement, decoherence, and noise.
- They are fundamental to quantum error correction and realistic quantum computing.

Concept Check

1. Why are quantum operations needed beyond unitary evolution?
2. What is the role of Kraus operators?
3. What condition must Kraus operators satisfy for a trace-preserving operation?
4. Name two practical applications of quantum operations.

Learning Outcomes

After completing this lecture, you will be able to

- Explain the role of single-qubit quantum gates.
- Write the matrix representations of the X, Y, Z and H gates.
- Determine the effect of these gates on computational basis states.
- Interpret these gates as rotations on the Bloch sphere.
- Recognize their importance in quantum circuits.

What are Single-Qubit Gates?

A quantum gate is a unitary operation that transforms a qubit.

For a single qubit,

$$|\psi'\rangle = U|\psi\rangle,$$

where

$$U^\dagger U = I.$$

Single-qubit gates manipulate the state of one qubit at a time and form the basic building blocks of quantum algorithms.

Pauli-X, Y and Z Gates

The Pauli gates are

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \qquad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Interpretation

- X: Bit-flip gate
- Y: Bit and phase flip
- Z: Phase-flip gate

Hadamard Gate

The Hadamard gate is

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Its action on the computational basis is

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}},$$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

The Hadamard gate creates equal superposition states.

Action on Basis States

Gate	$\|\mathbf{0}\rangle$	$\|\mathbf{1}\rangle$
X	$\|1\rangle$	$\|0\rangle$
Y	$i\|1\rangle$	$-i\|0\rangle$
Z	$\|0\rangle$	$-\|1\rangle$
H	$\frac{\|0\rangle + \|1\rangle}{\sqrt{2}}$	$\frac{\|0\rangle - \|1\rangle}{\sqrt{2}}$

Bloch Sphere Interpretation

Each gate corresponds to a rotation of the qubit state on the Bloch sphere.

- X: Rotation about the x-axis by 180°
- Y: Rotation about the y-axis by 180°
- Z: Rotation about the z-axis by 180°
- H: Maps computational basis states to superposition states

This geometric picture helps visualize quantum state transformations.

Applications

Single-qubit gates are used in

- State preparation
- Quantum algorithms
- Quantum error correction
- Quantum teleportation
- Quantum cryptography
- Universal quantum computation

Every quantum circuit begins with combinations of single-qubit operations.

Worked Example

Consider the qubit

$$|0\rangle.$$

Apply the Hadamard gate.

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Result:
The qubit is transformed from a basis state into an equal superposition of $|0\rangle$ and $|1\rangle$.

Summary and Concept Check

Summary

- Single-qubit gates perform unitary transformations on individual qubits.
- The Pauli gates X, Y, and Z implement bit, phase, and combined transformations.
- The Hadamard gate creates superposition states.
- These gates can be interpreted as rotations on the Bloch sphere.
- Single-qubit gates are fundamental components of all quantum circuits.

Concept Check

1. Which single-qubit gate acts as a quantum NOT gate?
2. Which gate creates an equal superposition from $|0\rangle$?
3. What is the effect of the Z gate on $|1\rangle$?
4. Why are single-qubit gates considered unitary operations?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the concept of quantum phase.
- Describe the S and T phase gates.
- Write the matrix representation of phase gates.
- Explain the effect of phase gates on qubit states.
- Recognize the importance of phase gates in quantum algorithms.

What is Quantum Phase?

A qubit may differ from another not only in its amplitudes but also in its relative phase.

A general qubit is

$$|\psi\rangle = \alpha|0\rangle + \beta e^{i\phi}|1\rangle.$$

The relative phase ϕ influences interference and measurement outcomes in quantum algorithms.

The S Gate

The S gate introduces a phase shift of

$$\frac{\pi}{2}.$$

Its matrix representation is

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}.$$

Its action is

$$S|0\rangle = |0\rangle, \qquad S|1\rangle = i|1\rangle.$$

The T Gate

The T gate introduces a phase shift of

$$\frac{\pi}{4}.$$

Its matrix representation is

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}.$$

Its action is

$$T|0\rangle = |0\rangle, \qquad T|1\rangle = e^{i\pi/4}|1\rangle.$$

General Phase Rotation Gate

A general phase rotation is represented by

$$R_\phi = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{bmatrix}.$$

Special cases include

$$\phi = \pi \Rightarrow Z,$$

$$\phi = \frac{\pi}{2} \Rightarrow S,$$

$$\phi = \frac{\pi}{4} \Rightarrow T.$$

Effect on the Bloch Sphere

Phase gates rotate the qubit state about the z-axis of the Bloch sphere.

Key points

- Probabilities remain unchanged.
- Relative phase changes.
- Interference effects are modified.
- Essential for many quantum algorithms.

Applications

Phase gates are widely used in

- Quantum Fourier Transform
- Quantum Phase Estimation
- Shor's Algorithm
- Quantum Error Correction
- Universal Quantum Gate Sets
- Fault-Tolerant Quantum Computing

Worked Example

Apply the S gate to the state

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Solution

$$S|\psi\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}.$$

The amplitudes remain unchanged, but the relative phase of $|1\rangle$ changes.

Summary and Concept Check

Summary

- Phase gates modify the relative phase of quantum states.
- The S gate applies a phase shift of $\pi/2$.
- The T gate applies a phase shift of $\pi/4$.
- Phase gates correspond to rotations about the z-axis of the Bloch sphere.
- They are fundamental to universal quantum computation and many quantum algorithms.

Concept Check

1. What is the purpose of a phase gate?
2. What phase shift does the S gate introduce?
3. How does the T gate differ from the S gate?
4. Why are phase gates important in quantum algorithms?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the operation of the CNOT gate.
- Write the matrix representation of the CNOT gate.
- Interpret control and target qubits.
- Describe common multi-qubit gates.
- Explain why CNOT is essential for quantum computation.

Why Multi-Qubit Gates?

Single-qubit gates manipulate individual qubits.

However, many quantum algorithms require interactions between qubits.

Multi-qubit gates enable

- Quantum correlations
- Entanglement
- Conditional operations
- Universal quantum computation

Controlled-NOT (CNOT) Gate

The CNOT gate has

- one **control** qubit,
- one **target** qubit.

Rule

- If the control qubit is $|0\rangle$, the target remains unchanged.
- If the control qubit is $|1\rangle$, the target is flipped.

The CNOT gate performs a conditional quantum NOT operation.

Truth Table and Matrix Representation

Truth Table

Input	Output
$\|00\rangle$	$\|00\rangle$
$\|01\rangle$	$\|01\rangle$
$\|10\rangle$	$\|11\rangle$
$\|11\rangle$	$\|10\rangle$

Matrix Representation

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Creating Entanglement

Start with

$$|00\rangle.$$

Apply

$$H$$

to the first qubit, then apply

$$\text{CNOT}.$$

The resulting state is

$$\boxed{\frac{|00\rangle + |11\rangle}{\sqrt{2}}}$$

which is a Bell state.

Other Multi-Qubit Gates

Common multi-qubit gates include

- Controlled-Z (CZ)
- SWAP Gate
- Toffoli Gate (CCNOT)
- Fredkin Gate (CSWAP)

These gates enable conditional logic, qubit exchange, and reversible computation.

Applications

Multi-qubit gates are essential for

- Quantum teleportation
- Superdense coding
- Quantum error correction
- Quantum Fourier Transform
- Shor's algorithm
- Grover's algorithm

Most practical quantum algorithms rely on CNOT and related controlled gates.

Worked Example

Apply the CNOT gate to

$$|10\rangle.$$

The first qubit is the control and is equal to $|1\rangle$.

Therefore, the target qubit is flipped:

$$|10\rangle \longrightarrow |11\rangle.$$

Summary and Concept Check

Summary

- Multi-qubit gates perform operations on two or more qubits.
- The CNOT gate flips the target qubit only when the control qubit is $|1\rangle$.
- CNOT is an entangling gate.
- Controlled gates are essential for universal quantum computation.
- Multi-qubit gates enable many quantum algorithms and protocols.

Concept Check

1. What are the roles of the control and target qubits in a CNOT gate?
2. What is the output of a CNOT gate for the input $|11\rangle$?
3. How can a Hadamard gate followed by a CNOT gate generate a Bell state?
4. Name two multi-qubit gates other than CNOT.

Learning Outcomes

After completing this lecture, you will be able to

- Explain the concept of quantum measurement.
- State the Born rule for measurement probabilities.
- Describe projective measurement.
- Explain wavefunction collapse.
- Determine measurement outcomes for simple qubit states.

What is Quantum Measurement?

Measurement is the process of extracting classical information from a quantum system.

Unlike classical measurement,

- the outcome is generally probabilistic,
- the quantum state changes after measurement.

Measurement is one of the fundamental postulates of quantum mechanics.

The Born Rule

Consider the qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

The probabilities of measurement are

$$P(0) = |\alpha|^2,$$

$$P(1) = |\beta|^2.$$

Since probabilities sum to one,

$$|\alpha|^2 + |\beta|^2 = 1.$$

Projective Measurement

The computational basis consists of

$$|0\rangle, \qquad |1\rangle.$$

Projective measurement projects the quantum state onto one of these basis states.

Possible outcomes are

$$|0\rangle \quad \text{or} \quad |1\rangle.$$

Wavefunction Collapse

Before measurement,
a qubit may exist in a superposition.

Example

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

After measurement,
the state collapses to either

$$|0\rangle$$

or

$$|1\rangle,$$

depending on the observed outcome.

Measurement in Quantum Circuits

Measurements usually appear at the end of a quantum circuit.

Purpose of measurement

- Read the computation result
- Convert a qubit into a classical bit
- Verify quantum algorithms

After measurement, the qubit no longer remains in its original superposition.

Applications

Quantum measurement is essential for

- Quantum algorithms
- Quantum teleportation
- Quantum key distribution
- Quantum error correction
- Quantum state tomography
- Quantum sensing

Worked Example

A qubit is prepared in

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Question:
What are the probabilities of obtaining

$$|0\rangle \quad \text{and} \quad |1\rangle?$$

Answer

$$P(0) = \frac{1}{2}, \qquad P(1) = \frac{1}{2}.$$

If the outcome is $|0\rangle$, the state collapses to $|0\rangle$.

Summary and Concept Check

Summary

- Quantum measurement extracts classical information from a quantum state.
- The Born rule determines measurement probabilities.
- Projective measurement yields one of the basis states.
- Measurement causes wavefunction collapse.
- Quantum measurement is fundamental to quantum algorithms and communication.

Concept Check

1. What is the Born rule?
2. What is meant by wavefunction collapse?
3. Why is quantum measurement probabilistic?
4. Why are measurements typically performed at the end of a quantum circuit?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the relationship between quantum gates and time evolution.
- Describe the role of the Hamiltonian in quantum dynamics.
- Write the time-evolution operator.
- Explain why quantum gates are unitary.
- Relate physical interactions to quantum gate operations.

Time Evolution in Quantum Mechanics

The evolution of a quantum state is governed by the time-dependent Schrödinger equation

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = H|\psi(t)\rangle,$$

where

- H is the Hamiltonian,
- $|\psi(t)\rangle$ is the quantum state.

The Hamiltonian determines how the system evolves with time.

The Time-Evolution Operator

For a time-independent Hamiltonian, the state evolves as

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle,$$

where

$$U(t) = e^{-iHt/\hbar}.$$

The operator $U(t)$ is called the **time-evolution operator**.

Quantum Gates as Unitary Operators

A quantum gate transforms a qubit according to

$$|\psi'\rangle = U|\psi\rangle.$$

Since

$$U = e^{-iHt/\hbar},$$

every ideal quantum gate is a unitary operation generated by a suitable Hamiltonian acting for a specified duration.

Thus, quantum gates are controlled time evolutions.

Physical Interpretation

A qubit interacts with external control fields such as

- Microwave pulses
- Laser pulses
- Magnetic fields
- Electric fields

These interactions define the Hamiltonian and determine which quantum gate is implemented.

Example: X Gate from Time Evolution

Consider a Hamiltonian proportional to the Pauli-X operator,

$$H \propto X.$$

Applying this Hamiltonian for an appropriate time produces a rotation about the x-axis of the Bloch sphere.

With the correct pulse duration, this evolution implements the quantum X (NOT) gate.

Applications

Viewing gates as time evolution is essential for

- Quantum hardware control
- Pulse-level programming
- Quantum simulations
- Superconducting qubits
- Trapped-ion quantum computers
- Quantum optimal control

Worked Example

A qubit is initially in the state

$$|0\rangle.$$

A control pulse generates a Hamiltonian that implements a Hadamard gate.

After the pulse,

$$|0\rangle \longrightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

The gate is therefore the result of controlled time evolution under the applied Hamiltonian.

Summary and Concept Check

Summary

- The Schrödinger equation governs quantum state evolution.
- The Hamiltonian determines the dynamics of a quantum system.
- The time-evolution operator is unitary.
- Ideal quantum gates correspond to controlled time evolution.
- Physical control pulses realize quantum gates in practical hardware.

Concept Check

1. What role does the Hamiltonian play in quantum evolution?
2. Why is the time-evolution operator unitary?
3. How are quantum gates related to the Schrödinger equation?
4. Name two physical control methods used to implement quantum gates.

Learning Outcomes

After completing this lecture, you will be able to

- Explain the motivation behind Bell's theorem.
- Describe local hidden-variable theories.
- Understand Bell measurements on entangled particles.
- Explain why Bell's theorem is important.
- Identify applications of Bell correlations.

Why Bell's Theorem?

Einstein, Podolsky and Rosen (EPR) argued that quantum mechanics might be incomplete.

They suggested that unknown hidden variables could determine measurement outcomes.

John Bell proposed a method to experimentally test this idea.

Bell's theorem compares the predictions of

- Local hidden-variable theories
- Quantum mechanics

Local Hidden Variables

A local hidden-variable theory assumes

- Measurement outcomes are predetermined.
- No information travels faster than light.
- Each particle carries complete local information.

Bell's theorem shows that these assumptions cannot reproduce all quantum predictions.

Bell Measurements

Two observers, traditionally called Alice and Bob,
measure entangled particles at different locations.

Each observer independently chooses a measurement basis.

The measurement outcomes are then compared to determine the strength of their correlations.

Bell States and Correlations

Consider the Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

If Alice measures

$$|0\rangle,$$

Bob is also found in

$$|0\rangle$$

when measured in the same basis.

These strong correlations agree with quantum mechanics.

Significance of Bell's Theorem

Bell's theorem demonstrates that

- Quantum mechanics predicts correlations beyond classical expectations.
- Local hidden-variable theories cannot explain all observed results.
- Entanglement is a genuine quantum resource.

Bell experiments strongly support the predictions of quantum mechanics.

Applications

Bell correlations are important in

- Quantum key distribution
- Quantum teleportation
- Device-independent quantum cryptography
- Quantum networking
- Quantum communication
- Tests of quantum foundations

Worked Example

Two qubits are prepared in the Bell state

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Question:
If Alice measures

$$|1\rangle,$$

what is Bob's result when measuring in the same basis?

Answer:
Bob also measures

$$|1\rangle.$$

This illustrates the strong correlations predicted by quantum mechanics.

Summary and Concept Check

Summary

- Bell's theorem tests whether local hidden-variable theories can explain quantum correlations.
- Bell measurements compare outcomes obtained by distant observers.
- Experimental results support quantum mechanics over local hidden-variable models.
- Bell correlations arise from entanglement.
- Bell's theorem underpins many modern quantum information protocols.

Concept Check

1. What question does Bell's theorem address?
2. What is meant by a local hidden-variable theory?
3. Why are Bell states useful in Bell experiments?
4. Name two applications that rely on Bell correlations.

Learning Outcomes

After completing this lecture, you will be able to

- Explain the purpose of a quantum circuit.
- Identify the basic components of a quantum circuit.
- Interpret common quantum circuit symbols.
- Describe the sequence of operations in a quantum circuit.
- Relate quantum circuits to quantum algorithms.

What is a Quantum Circuit?

A quantum circuit is a graphical representation of a quantum computation.

A circuit consists of

- Qubit wires
- Quantum gates
- Measurement operations
- Classical output bits

Quantum circuits provide a convenient way to design and analyze quantum algorithms.

Basic Components

A typical quantum circuit contains

- Input qubits initialized to known states
- Single-qubit gates (X, Y, Z, H, S, T)
- Multi-qubit gates (CNOT, CZ, SWAP)
- Measurement operations
- Classical registers storing the outcomes

Information flows from left to right.

Common Circuit Symbols

Symbol	**Meaning**
Horizontal line	Qubit wire
Rectangle labelled H, X, Z	Single-qubit gate
$\bullet$	Control qubit
$\oplus$	Target of a CNOT gate
Meter symbol	Measurement
Double line	Classical bit

Execution of a Quantum Circuit

The execution proceeds in the following order:

1. Initialize qubits.
2. Apply quantum gates.
3. Create superposition and entanglement (if required).
4. Perform measurement.
5. Obtain classical output.

Each gate represents a unitary transformation applied to the quantum state.

Example Circuit

A simple circuit for creating a Bell state:

$$|00\rangle$$

1. Apply a Hadamard gate to the first qubit.
2. Apply a CNOT gate.
3. Measure both qubits.

The circuit produces correlated measurement outcomes characteristic of an entangled state.

Applications of Quantum Circuits

Quantum circuits are used to implement

- Quantum algorithms
- Quantum error correction
- Quantum communication protocols
- Quantum simulations
- Quantum machine learning
- Cryptographic protocols

Nearly all gate-based quantum computers are programmed using quantum circuits.

Worked Example

Consider the following sequence:

$$|0\rangle \rightarrow H \rightarrow X \rightarrow M$$

where M denotes measurement.

Discussion:

- The Hadamard gate creates a superposition.
- The X gate exchanges the amplitudes of the basis states.
- Measurement produces a classical bit according to the final state's probabilities.

Summary and Concept Check

Summary

- Quantum circuits describe quantum computations graphically.
- A circuit consists of qubits, gates, and measurements.
- Gates perform unitary operations on qubits.
- Measurements convert quantum information into classical information.
- Quantum circuits form the basis of quantum programming and algorithms.

Concept Check

1. What are the main components of a quantum circuit?
2. What is the purpose of a qubit wire?
3. Why are measurements usually placed at the end of a circuit?
4. Which gate is commonly used to create entanglement when combined with a Hadamard gate?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the objective of the Deutsch algorithm.
- Distinguish between constant and balanced Boolean functions.
- Describe the role of the quantum oracle.
- Explain how superposition enables quantum parallelism.
- Interpret the measurement outcome of the algorithm.

Problem Statement

Consider a Boolean function

$$f : \{0, 1\} \to \{0, 1\}.$$

The function is guaranteed to be either

- Constant: $f(0) = f(1)$
- Balanced: $f(0) \neq f(1)$

Goal:
Determine whether f is constant or balanced using as few function evaluations as possible.

Classical vs Quantum Solution

Classical Computer	**Quantum Computer**
Needs two evaluations	Needs one oracle query
Checks $f(0)$ and $f(1)$	Evaluates both inputs simultaneously through superposition
Deterministic	Uses interference to reveal the answer

The Deutsch algorithm demonstrates the power of quantum computation.

Quantum Circuit

The Deutsch algorithm uses

- Two qubits
- Hadamard gates
- Quantum oracle U_f
- Final measurement

Circuit sequence:

$$|0\rangle|1\rangle \rightarrow H \rightarrow U_f \rightarrow H \rightarrow \text{Measure}$$

Role of the Oracle

The oracle implements the function $f(x)$.

Its action is

$$U_f|x, y\rangle = |x,\ y \oplus f(x)\rangle,$$

where $\oplus$ denotes XOR.

The oracle is treated as a black box whose internal implementation is unknown.

Quantum Parallelism

After the initial Hadamard gates,
the first qubit is in a superposition of

$$|0\rangle \quad \text{and} \quad |1\rangle.$$

The oracle acts on both components simultaneously.

Quantum interference then amplifies the correct outcome before measurement.

Measurement Outcome

After the final Hadamard gate,
only the first qubit is measured.

Measurement result

- $|0\rangle \Rightarrow$ Constant function
- $|1\rangle \Rightarrow$ Balanced function

Only one oracle query is required.

Worked Example

Suppose

$$f(0) = 0, \qquad f(1) = 1.$$

Since the outputs are different,
the function is balanced.

The Deutsch algorithm measures

$$|1\rangle$$

on the first qubit, correctly identifying the function with a single oracle evaluation.

Summary and Concept Check

Summary

- The Deutsch algorithm determines whether a Boolean function is constant or balanced.
- Quantum superposition enables simultaneous evaluation of multiple inputs.
- The oracle encodes the unknown function.
- Quantum interference extracts the desired information.
- Only one oracle query is required.

Concept Check

1. What is the objective of the Deutsch algorithm?
2. What is the role of the oracle U_f?
3. Why does the quantum algorithm require only one oracle query?
4. What does the measurement outcome $|1\rangle$ indicate?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the purpose of the Deutsch–Jozsa algorithm.
- Distinguish between constant and balanced Boolean functions.
- Describe the quantum circuit of the algorithm.
- Explain the role of quantum interference.
- Compare the quantum and classical computational costs.

Problem Statement

Consider a Boolean function

$$f : \{0,1\}^n \to \{0,1\},$$

which is guaranteed to be either

- **Constant**: the output is the same for every input.
- **Balanced**: exactly half of the inputs map to 0 and half to 1.

Goal:
Determine whether f is constant or balanced using the fewest possible oracle queries.

Classical vs Quantum Approach

Classical Algorithm	**Deutsch–Jozsa Algorithm**
May require up to $2^{n-1}+1$ oracle evaluations (deterministically)	Requires exactly one oracle query
Checks many inputs	Processes all inputs in superposition
Sequential evaluation	Quantum interference reveals the answer

Quantum Circuit

The Deutsch–Jozsa circuit consists of

- n input qubits
- One auxiliary qubit
- Hadamard gates
- Oracle U_f
- Final Hadamard gates
- Measurement of the input register

General sequence:

$$|0\rangle^{\otimes n}|1\rangle \rightarrow H \rightarrow U_f \rightarrow H \rightarrow \text{Measure}$$

Role of the Oracle

The oracle performs the transformation

$$U_f|x, y\rangle = |x,\ y \oplus f(x)\rangle.$$

The oracle evaluates the function for every basis state present in the superposition.

Quantum interference encodes the global property of the function into the amplitudes of the input register.

Quantum Interference

After the oracle,
the second set of Hadamard gates causes constructive or destructive interference.

Outcome:

- Constant function: all amplitudes interfere constructively toward $|00\cdots0\rangle$.
- Balanced function: interference cancels the amplitude of $|00\cdots0\rangle$.

Measurement Rule

Measure only the input qubits.

Interpretation:

- If the outcome is

$$|00\cdots0\rangle,$$

  the function is constant.

- If any other bit string is obtained, the function is balanced.

Only one oracle query is required.

Worked Example

Suppose

$$n = 2,$$

and

$$f(00) = 0, \quad f(01) = 0, \quad f(10) = 1, \quad f(11) = 1.$$

Exactly half of the outputs are 0 and half are 1.

Therefore, the function is balanced.

The Deutsch–Jozsa algorithm identifies this using a single oracle evaluation.

Summary and Concept Check

Summary

- The Deutsch–Jozsa algorithm generalizes the Deutsch algorithm to n-bit Boolean functions.
- The oracle acts on all inputs simultaneously through superposition.
- Quantum interference reveals whether the function is constant or balanced.
- A deterministic classical algorithm may require exponentially many oracle evaluations.
- The quantum algorithm requires only one oracle query.

Concept Check

1. What is the promise made about the Boolean function?
2. What is the purpose of the oracle U_f?
3. Why is quantum interference essential to the algorithm?
4. What measurement result indicates a constant function?

Learning Outcomes

After completing this lecture, you will be able to

- Explain the purpose of Grover's algorithm.
- Describe the role of the oracle.
- Explain amplitude amplification.
- Compare classical and quantum search complexities.
- Identify practical applications of Grover's algorithm.

The Search Problem

Suppose a database contains

$$N$$

unsorted items.

Goal:
Find one marked item.

Examples include

- Searching an unsorted database
- Password recovery
- Pattern matching
- Constraint satisfaction

Classical vs Quantum Search

Classical Search	**Grover's Algorithm**
Average complexity:	Average complexity:
$O(N)$	$O(\sqrt{N})$
Sequential checking	Amplitude amplification
Linear scaling	Quadratic speedup

Main Steps of Grover's Algorithm

The algorithm consists of

1. Initialize all qubits.
2. Apply Hadamard gates.
3. Apply the oracle.
4. Apply the diffusion operator.
5. Repeat the last two steps approximately

$$\frac{\pi}{4}\sqrt{N}$$

   times.
6. Measure the qubits.

The Oracle

The oracle identifies the desired solution.

It changes the phase of the marked state.

Conceptually,

$$|x\rangle \rightarrow -|x\rangle$$

only for the correct answer.

All other states remain unchanged.

Amplitude Amplification

After the oracle,
the diffusion operator increases the probability amplitude of the marked state.

Repeated applications

- amplify the correct state,
- suppress the remaining states.

This process is called **amplitude amplification**.

Applications

Grover's algorithm can be applied to

- Database search
- Optimization
- Cryptanalysis
- Artificial intelligence
- Machine learning
- Pattern recognition

It provides a quadratic speedup for many unstructured search problems.

Worked Example

Suppose a database contains

$$16$$

items.

Classical search requires approximately

$$16$$

queries in the worst case.

Grover's algorithm requires approximately

$$\sqrt{16} = 4$$

oracle iterations.

The search is therefore significantly faster.

Summary and Concept Check

Summary

- Grover's algorithm searches an unstructured database.
- The oracle marks the desired state.
- The diffusion operator amplifies the probability of the marked state.
- The algorithm achieves a quadratic speedup.
- Grover's algorithm has applications in search and optimization.

Concept Check

1. What is the purpose of the oracle in Grover's algorithm?
2. What is amplitude amplification?
3. How does the complexity of Grover's algorithm compare with classical search?
4. Name two applications of Grover's algorithm.

Learning Outcomes

After completing this lecture, you will be able to

- Explain the purpose of Shor's algorithm.
- Describe the integer factorization problem.
- Understand the concept of period finding.
- Explain why the Quantum Fourier Transform is important.
- Recognize the impact of Shor's algorithm on cryptography.

The Integer Factorization Problem

Given a composite integer

$$N,$$

determine its prime factors.

Examples

$$15 = 3 \times 5,$$

$$21 = 3 \times 7.$$

For very large integers, factorization becomes computationally difficult for classical computers.

Why is Factorization Important?

Modern public-key cryptography, including RSA, relies on the practical difficulty of factoring large integers.

If large numbers can be factored efficiently,

- encrypted messages may be decrypted,
- digital signatures may be compromised.

Shor's algorithm demonstrates that quantum computers can efficiently solve the factorization problem.

Main Steps of Shor's Algorithm

The algorithm proceeds through the following stages.

1. Select a random integer.
2. Perform modular exponentiation.
3. Find the period of the function using a quantum computer.
4. Use the period to determine the factors.
5. Verify the result classically.

The quantum speedup comes from the period-finding step.

Period Finding

Consider the function

$$f(x) = a^x \bmod N.$$

The function repeats after some integer r,
called the period.

Finding this period efficiently is difficult for classical computers but can be achieved using a quantum computer.

Role of the Quantum Fourier Transform

The Quantum Fourier Transform (QFT) is used to extract the period of the modular function.

The QFT

- transforms quantum amplitudes,
- enhances periodic information,
- enables efficient period estimation.

The QFT is the key quantum component of Shor's algorithm.

Classical vs Quantum Complexity

Classical Methods	**Shor's Algorithm**
Require very large computation times for large integers	Runs in polynomial time with a sufficiently capable quantum computer
Security basis of RSA	Challenges RSA security

Shor's algorithm illustrates a significant quantum computational advantage.

Worked Example

Factor the integer

$$15.$$

Shor's algorithm determines the period of an appropriate modular function.

The period is then used to compute

$$3 \quad \text{and} \quad 5,$$

the prime factors of 15.

The detailed arithmetic is omitted here; the focus is on the algorithmic workflow.

Summary and Concept Check

Summary

- Shor's algorithm solves the integer factorization problem.
- It combines classical computation with quantum period finding.
- The Quantum Fourier Transform enables efficient period estimation.
- The algorithm offers a polynomial-time quantum solution to integer factorization.
- Shor's algorithm has profound implications for modern cryptography.

Concept Check

1. What computational problem does Shor's algorithm solve?
2. Why is period finding central to the algorithm?
3. What is the role of the Quantum Fourier Transform?
4. Why is Shor's algorithm important for RSA cryptography?

Tutorial Objectives

After this tutorial you should be able to

- Recall all four Bell states.
- Identify entangled and separable states.
- Predict measurement outcomes.
- Construct Bell states using quantum gates.
- Explain why Bell states are useful in quantum information.

Quick Revision

Write the four Bell states.

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

Practice 1: Identify the State

Determine whether each state is

- Product (Separable)
- Entangled

$$|00\rangle$$

$$|0\rangle \otimes |1\rangle$$

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$\frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

Explain your reasoning.

Practice 2: Measurement Outcomes

Suppose two qubits are prepared in

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Predict the outcome if

1. Alice measures $|0\rangle$.
2. Alice measures $|1\rangle$.
3. Bob measures immediately afterwards.

Discuss the observed correlations.

Practice 3: Construct a Bell State

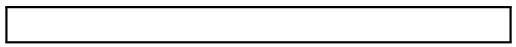
Initial state

$$|00\rangle$$

Apply the following operations

1. Hadamard gate on the first qubit.
2. CNOT gate.

Question:
Which Bell state is produced?



Worked Example

Starting from

$$|00\rangle$$

After applying

$$H$$

to the first qubit,

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

Applying CNOT gives

$$\boxed{\frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\Phi^+\rangle}$$

Applications of Bell States

Bell states are fundamental to

- Quantum teleportation
- Superdense coding
- Quantum key distribution
- Bell inequality experiments
- Quantum networking
- Quantum repeaters

Discuss why entanglement is essential in each application.

Multiple Choice Questions

1. Which Bell state is produced by

$$H + \mathrm{CNOT}$$

acting on $|00\rangle$?

- A. $|\Phi^+\rangle$
- B. $|\Phi^-\rangle$
- C. $|\Psi^+\rangle$
- D. $|\Psi^-\rangle$

2. Bell states are

- A. Product states
- B. Maximally entangled states
- C. Classical states
- D. Mixed states

Summary

Today you practised

- Bell state identification
- Measurement correlations
- Entanglement
- Bell-state preparation
- Applications of Bell states

Self-Study

- Memorize the four Bell states.
- Derive Bell states using the H and CNOT gates.
- Explain how Bell states are used in quantum teleportation.

Tutorial Objectives

After completing this tutorial, you should be able to

- Predict the action of common quantum gates.
- Analyze simple quantum circuits.
- Distinguish between bit-flip and phase-flip operations.
- Determine measurement outcomes.
- Construct Bell states using elementary gates.

Quick Revision

Complete the following table.

Gate	**Action on $\|0\rangle$**	**Action on $\|1\rangle$**
X	?	?
Y	?	?
Z	?	?
H	?	?
S	?	?
T	?	?

Practice 1: Single-Qubit Gates

Determine the output state for each operation.

1. $X|0\rangle$
2. $Z|1\rangle$
3. $H|0\rangle$
4. $H|1\rangle$
5. $S|1\rangle$
6. $T|1\rangle$

Discuss whether each gate changes

- the bit value,
- the phase,
- or both.

Practice 2: Circuit Analysis

Consider the circuit

$$|0\rangle \rightarrow H \rightarrow X \rightarrow H \rightarrow M$$

where M denotes measurement.

Questions

- What is the quantum state after each gate?
- What measurement outcome is expected?

Practice 3: CNOT Gate

Determine the output for the following inputs.

Input	Output
$\|00\rangle$	?
$\|01\rangle$	?
$\|10\rangle$	?
$\|11\rangle$	?

Explain the roles of the control and target qubits.

Worked Example

Initial state

$$|00\rangle.$$

Apply

$$H$$

to the first qubit.

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

Apply CNOT.

$$\boxed{\frac{|00\rangle + |11\rangle}{\sqrt{2}}}$$

The circuit prepares the Bell state $|\Phi^+\rangle$.

Practice 4: Measurement

A qubit is prepared in

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Questions

1. What is the probability of measuring $|0\rangle$?
2. What is the probability of measuring $|1\rangle$?
3. What is the state after measuring $|0\rangle$?

Use the Born rule to justify your answers.

Multiple Choice Questions

1. Which gate creates an equal superposition from $|0\rangle$?

- A. X
- B. Z
- C. H
- D. S

2. Which gate can create entanglement when combined with a Hadamard gate?

- A. Z
- B. T
- C. CNOT
- D. S

Summary

Today you practised

- Single-qubit gate analysis
- Phase-gate operations
- CNOT truth table
- Circuit analysis
- Bell-state preparation
- Quantum measurement

Suggested Self-Study

- Memorize the matrices of the X, Y, Z, H, S and T gates.
- Analyze simple two-qubit circuits using H and CNOT.
- Predict measurement outcomes for superposition states.

Tutorial Objectives

After completing this tutorial, you should be able to

- Compare major quantum algorithms.
- Identify the computational problem solved by each algorithm.
- Explain the role of the oracle.
- Compare classical and quantum complexities.
- Select an appropriate algorithm for a given problem.

Quick Revision

Complete the following table.

Algorithm	**Problem Solved**	**Speedup**
Deutsch	?	?
Deutsch–Jozsa	?	?
Grover	?	?
Shor	?	?

Practice 1: Match the Algorithm

Match each problem with the correct algorithm.

A. Integer factorization	1. Grover
B. Database search	2. Shor
C. Constant vs balanced function	3. Deutsch–Jozsa
D. Single-bit Boolean function	4. Deutsch

Discuss your answers before revealing the solution.

Practice 2: Complexity Comparison

Fill in the missing entries.

Algorithm	Classical	Quantum
Deutsch	?	1 query
Deutsch–Jozsa	?	1 query
Grover	$O(N)$	?
Shor	Very difficult for large integers	Polynomial time

Explain the meaning of the improvement for each algorithm.

Practice 3: Oracle Analysis

Answer the following questions.

1. Which algorithms use a quantum oracle?
2. What information does the oracle provide?
3. Does the oracle solve the problem directly?
4. Why is interference required after the oracle?

Discuss how the oracle differs from a conventional subroutine.

Worked Example

A company needs to search an unsorted database containing one million records.

Question:
Which quantum algorithm is most suitable?

Answer
Grover's algorithm.

Reason:
It performs unstructured search in approximately

$$O(\sqrt{N})$$

queries, providing a quadratic speedup over classical search.

Practice 4: Think and Discuss

Discuss the following questions.

1. Why does Grover's algorithm provide only a quadratic speedup?
2. Why is Shor's algorithm important for RSA cryptography?
3. Why do Deutsch and Deutsch–Jozsa algorithms use Hadamard gates?
4. Which algorithm demonstrates the largest known computational advantage?

Multiple Choice Questions

1. Which algorithm is used for integer factorization?

- A. Deutsch
- B. Grover
- C. Shor
- D. Deutsch–Jozsa

2. Which algorithm searches an unstructured database?

- A. Grover
- B. Shor
- C. Deutsch
- D. Simon

Summary

Today you practised

- Comparing quantum algorithms
- Identifying computational problems
- Understanding oracle-based computation
- Comparing classical and quantum complexity
- Selecting suitable algorithms for applications

Suggested Self-Study

- Draw a comparison chart of Deutsch, Deutsch–Jozsa, Grover, and Shor algorithms.
- Summarize the key idea behind each algorithm in one sentence.
- Identify one real-world application for each algorithm.

Tutorial Objectives

After this tutorial you should be able to

- Create simple quantum circuits.
- Apply single- and two-qubit gates.
- Perform quantum measurements.
- Predict circuit outputs.
- Relate quantum circuits to program code.

Quick Revision

Complete the table.

Operation	Qiskit Method
Hadamard Gate	?
Pauli-X Gate	?
CNOT Gate	?
Measurement	?

Expected answers:
`h()`, `x()`, `cx()`, `measure()`.

Practice 1: Create a Circuit

Write a program to create a one-qubit circuit.

```
from qiskit import QuantumCir

qc = QuantumCircuit(1,1)

qc.h(0)
qc.measure(0,0)
```

Question:
What measurement outcomes are expected?

Practice 2: Bell-State Circuit

Complete the missing statement.

```
qc = QuantumCircuit(2,2)

qc.h(0)

# Missing line

qc.measure([0,1],[0,1])
```

Which instruction prepares a Bell state?

`qc.cx(0,1)`

Practice 3: Circuit Analysis

Determine the output of the circuit

$$|0\rangle \rightarrow H \rightarrow Z \rightarrow H \rightarrow M$$

Questions

- What is the state after each gate?
- What is the expected measurement?
- Which single-qubit gate is equivalent to this sequence?

Worked Example

Initial state

$$|00\rangle$$

Apply

$$H$$

to the first qubit.

Then apply

$$\text{CNOT}.$$

Final state

$$\boxed{\frac{|00\rangle + |11\rangle}{\sqrt{2}}}$$

This circuit prepares the Bell state $|\Phi^+\rangle$

Practice 4: Debug the Circuit

Identify the mistakes.

1. Measuring before applying all gates.
2. Incorrect control-target order in CNOT.
3. Missing classical register.
4. Applying gates to invalid qubit indices.

Discuss how each error affects the circuit.

Multiple Choice Questions

1. Which command applies a Hadamard gate?

- A. `qc.h(0)`
- B. `qc.x(0)`
- C. `qc.cx(0,1)`
- D. `qc.measure()`

2. Which gate creates entanglement with a Hadamard gate?

- A. Z
- B. X
- C. CNOT
- D. T

Summary

Today you practised

- Writing simple quantum programs
- Creating quantum circuits
- Preparing Bell states
- Measuring qubits
- Debugging quantum circuits

Suggested Self-Study

- Create a circuit for each Bell state.
- Simulate the Deutsch algorithm in Qiskit.
- Modify Grover's algorithm for a four-element search space.